CHAPTER I

INTRODUCTION

Project Context

In an era of rapid technological advancement, the demand for sustainable and energy-efficient solutions has become a global priority. One of the most significant contributors to unnecessary energy consumption is inefficient lighting systems in residential, commercial, and public spaces. Many establishments continue to rely on conventional lighting methods, where lights remain switched on regardless of natural daylight availability or human presence. This results in excessive power usage, increased electricity costs, and negative environmental impacts.

With the emergence of the Internet of Things (IoT), automation and smart control have transformed traditional systems into intelligent, responsive technologies. IoT enables interconnected devices to communicate, collect, and exchange real-time data, creating opportunities to monitor and optimize energy consumption. The integration of IoT in lighting systems allows users to control lights remotely, automate functions based on motion or ambient light, and track usage patterns through a centralized dashboard or mobile application.

The IoT-Based Smart Lighting and Energy Saving System addresses the growing need for intelligent energy management. By using sensors, microcontrollers, and IoT platforms, the system automates lighting operations to ensure that lights are only active when necessary. For instance, during the daytime, the system detects sufficient natural light and automatically turns off artificial lights. At night, it activates lights only when motion is detected, ensuring both convenience and safety. Users can also manage and customize settings through a mobile app or IoT dashboard.

This project promotes energy conservation and environmental sustainability while providing a modern, cost-efficient lighting solution. It aligns with the principles of smart home and smart city development, where technology enhances comfort, efficiency, and resource management. Ultimately, the IoT-Based Smart Lighting and Energy Saving System demonstrates how innovation can create meaningful change toward a more sustainable future.

Objectives of the Study

The general objective of this study is to design and develop an IoT-Based Smart Lighting and Energy Saving System that automates lighting operations to minimize energy waste and enhance user convenience.

Specifically, this study aims to:

1. Develop a system that can monitor lighting usage in real-time using light and motion sensors.
2. Implement an automated lighting mechanism that adjusts based on ambient light intensity or detected movement.
3. Integrate a mobile application or IoT dashboard that allows users to remotely control and customize lighting settings.

Scope and Limitations of the Study

The scope of the "sLITES: An IoT-Based Smart Lighting Intelligence for Targeted Energy Saving" centers on developing an automated lighting solution that intelligently manages power usage through sensor-based control and IoT integration. The system is designed to monitor lighting conditions in real-time, automate the switching of lights based on environmental brightness and motion detection, and allow remote control through a mobile application or IoT dashboard. Core features include real-time light intensity monitoring using LDR sensors, motion-based automation through PIR sensors, microcontroller-based processing and decision-making, and remote management via an IoT platform. This integrated system aims to optimize energy consumption, reduce electricity costs, and promote sustainability by ensuring that lights operate only when necessary.

However, the system is developed primarily as a prototype for indoor lighting automation and may require further calibration for outdoor or large-scale commercial use. The accuracy of motion and light detection depends on sensor placement, environmental factors, and hardware quality, which may influence system performance. The mobile application or IoT dashboard focuses on basic control and monitoring functions and may not include advanced features such as voice command integration, adaptive learning, or multi-device synchronization. Additionally, the system's operation relies on stable internet connectivity; interruptions or weak connections may affect remote control capabilities. Future expansions in IoT technology or changes in hardware compatibility may also impact performance and scalability.

Definition of Terms

To ensure clarity and understanding, the following terms are defined as they are used in this study:

Automation – The process of using technology to perform tasks automatically without human intervention.

Energy Efficiency – The ability to use less energy to perform the same task, thereby reducing waste and saving costs.

Internet of Things (IoT) – A network of interconnected devices that communicate and exchange data over the internet.

Light Dependent Resistor (LDR) – A type of sensor that detects light intensity; its resistance changes based on the amount of light it receives.

Microcontroller – A compact integrated circuit that serves as the brain of the system, processing sensor data and controlling outputs.

Motion Sensor (PIR) – A sensor that detects movement within its range, triggering the system to respond (e.g., turning on a light).

Real-Time Monitoring – The continuous observation of system activity, such as lighting status and energy usage, as it happens.

Smart Bulb/LED – A light source that can be controlled digitally, often supporting remote operation or automation.

Sustainability – The practice of meeting present energy needs without compromising the ability of future generations to meet theirs.

User Interface (UI) – The platform, such as a mobile app or dashboard, that allows users to interact with and control the system.

CHAPTER 2
REQUIREMENTS SPECIFICATION

Hardware and Software Requirements

The IoT-Based Smart Lighting and Energy Saving System requires a combination of hardware and software components that work together to monitor light intensity, detect motion, and automate lighting operations. These components ensure efficient energy usage and remote accessibility through an IoT-based interface.

Table 1. Hardware Requirements

Component	Description
Orange Pi (Microcontroller)	Serves as the main control unit that processes input from sensors and manages lighting automation. It supports GPIO connections, Wi-Fi communication, and cloud integration.
Light Dependent Resistor (LDR)	Detects ambient light intensity to determine when lights should be turned ON or OFF.

Passive Infrared Sensor (PIR)	Detects motion within its range to trigger the light activation process.
Relay Module	Acts as a switch that controls the electrical flow to the lights based on Orange Pi commands.
Smart LED Bulbs / Lights	Serve as the system's output, providing illumination that can be automatically or manually controlled.
Wi-Fi Adapter (if not built-in)	Allows the Orange Pi to communicate with the IoT platform over a wireless network.
Power Supply (optional)	Provides stable voltage and current required by Orange Pi and sensors.

The hardware components utilized in the intelligent lighting system are summarized in Table 1. Each component was selected for its specific role in creating a reliable, automated, and energy-efficient solution. The integration of these elements forms a

cohesive IoT framework capable of real-time environmental sensing, local processing, and cloud-based control.

Table 2. Softwaare Requirements

Component	Purpose / Description
Python Programming Language	Used to write and execute the automation scripts on the Orange Pi.
IoT Platform (e.g., Blynk, Firebase, or MQTT Broker)	Handles real-time communication between hardware and the user interface.
Mobile Application or Web Dashboard	Allows users to monitor lighting status, view sensor data, and control the system remotely.
Operating System (Ubuntu for Orange Pi)	Enables system operation and supports IoT software integration.
Web Browser	Used to access the dashboard for data visualization and remote control.

The software stack for the smart lighting system is designed to enable local automation, remote control, and real-time monitoring. Each layer—from the operating system to the user interface—plays a critical role in delivering a seamless and intelligent lighting experience.

Functional Requirements

This section outlines the core functions of the Smart Lighting Automation System, including ambient light sensing, motion detection, automated control, remote access, and manual override. These functionalities ensure that the system meets the operational demands of convenience, energy efficiency, and user accessibility.

Table 3. Funtional Requirements

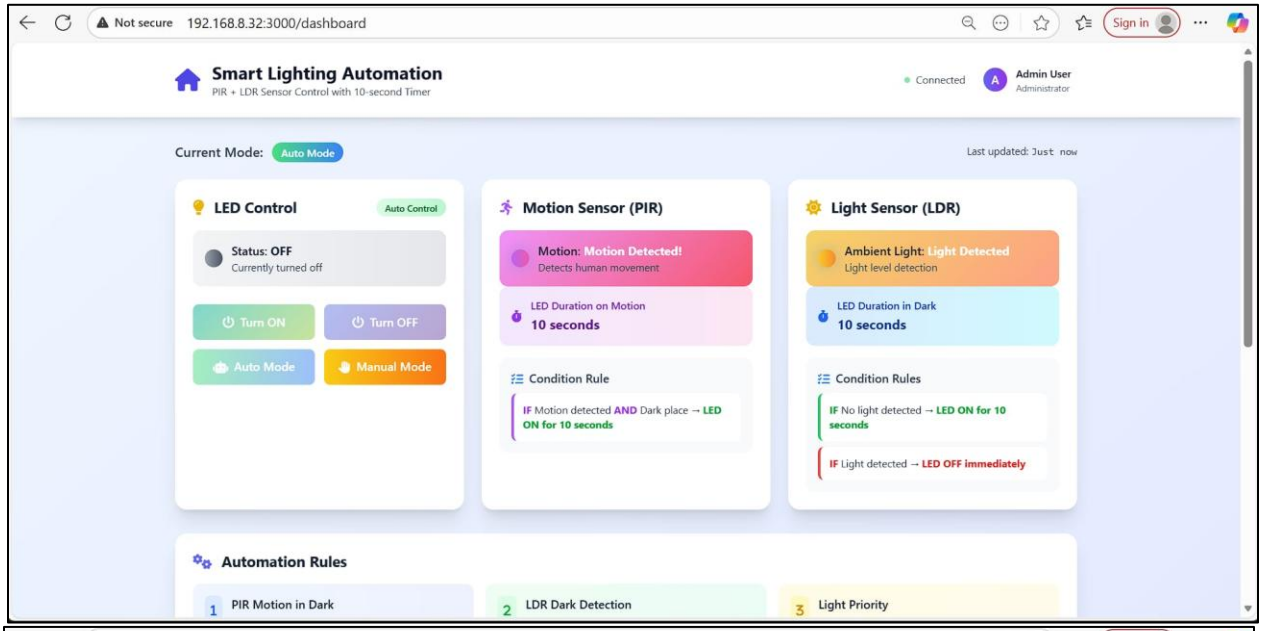
Feature	Description
Ambient Light Detection	The system automatically measures ambient light levels using an LDR sensor to determine if lighting is necessary, enabling energy-efficient daylight-based operation.
Motion-Triggered Activation	A PIR sensor detects human presence within its range, triggering the lighting automation process only when motion is sensed, enhancing convenience and energy savings.
Intelligent Automation Logic	The Orange Pi executes Python-based logic to integrate sensor inputs (LDR and PIR), determining when to turn

	lights ON/OFF based on ambient light and motion conditions.
Remote Access & Control	Users can monitor lighting status, view real-time sensor data, and control the system manually via a mobile app or web dashboard, regardless of location.
Manual Override Option	The system supports manual control through the user interface, allowing users to temporarily bypass automation and set lights to their preferred state.

Table 3 presents the key features of the Smart Lighting Automation System, outlining its core capabilities from sensing to user control. These six integrated features enable a fully automated, energy-efficient, and user-adaptable lighting solution. The system combines local sensor intelligence with remote accessibility and cloud-based oversight to deliver reliable, convenient, and scalable smart lighting management.

User Interface

- The system must provide a graphical dashboard or mobile app for controlling and monitoring lighting in real time.
- Users can manually switch lights ON/OFF or set automation rules.
- The interface should display live sensor readings, such as light intensity and motion detection.
- The user can customize automation thresholds and lighting schedules through the interface.



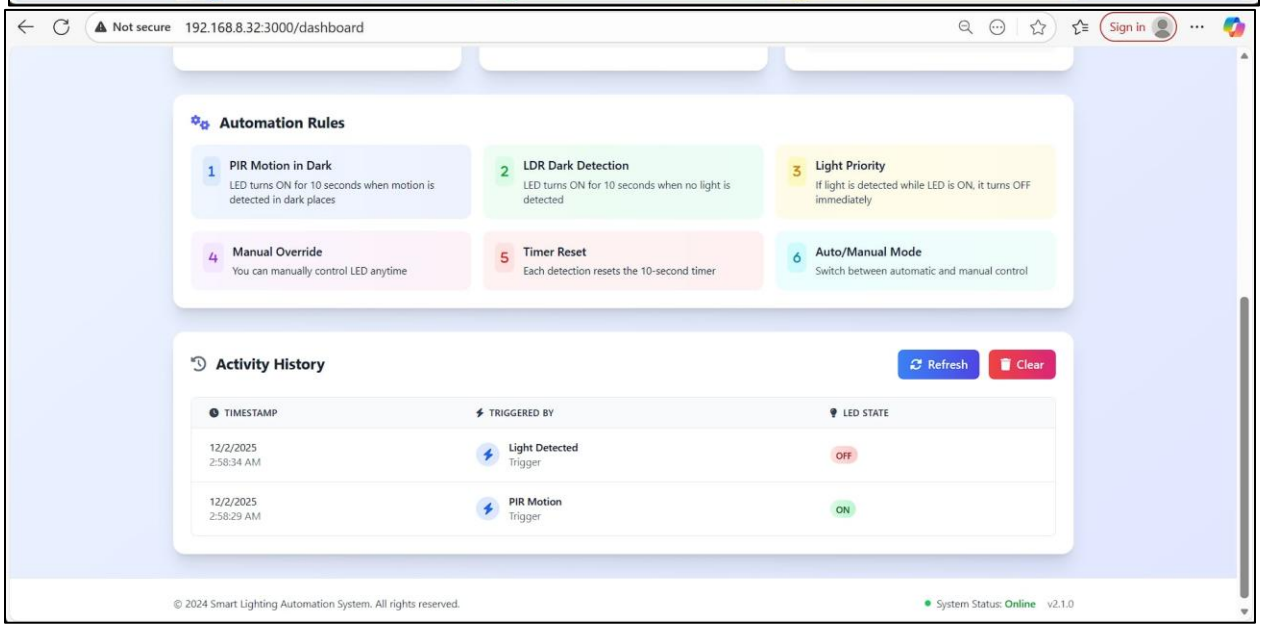


Figure 1 & 2. Dashboard

Hardware Interface

- The LDR sends real-time light level data to the Orange Pi via analog/digital GPIO pins.
- The PIR sensor detects motion and triggers signals to the Orange Pi.

- The Relay Module receives control signals from the Orange Pi to switch lights accordingly.
- The system connects to the IoT platform via Wi-Fi for data exchange.

Software Interface

- The Orange Pi communicates with the IoT platform using HTTP, MQTT, or WebSocket protocols.
- Data from sensors are uploaded to the IoT cloud and visualized on the mobile/web dashboard.
- Commands from the user interface are sent back to the Orange Pi to update device states.
- The system ensures synchronization between the local device and cloud server.

Security Interface

- Only registered and authenticated users can access the control dashboard or mobile app.
- The system uses secure communication protocols (SSL/TLS) for data transmission.
- Login credentials (username and password) are required to access and control the system.
- The Orange Pi runs with restricted permissions to prevent unauthorized system modification.

Technical Background

Hardware Specifications

Table 4.Laptop Specifications

Component	**Specification / Description**
Microcontroller	Orange Pi (Quad-core ARM CPU, built-in Wi-Fi/Ethernet, GPIO interface)
Light Sensor (LDR)	10–20kΩ light-dependent resistor for brightness detection
PIR Sensor	Detection range: 5–7 meters, 120° field of view
Relay Module	5V single-channel relay module for switching AC lights
Smart LED Light	220V LED bulb compatible with relay switching
Power Supply(optional)	5V–2A regulated power supply for Orange Pi and sensors
Wi-Fi Adapter	2.4 GHz wireless connection

	(built-in or USB module)

Software Specifications

Table 5. Software Specifications

Component	**Specification / Description**
Python/Node js	Programming language for automation and sensor data processing
Ubuntu OS	Operating system installed on the Orange Pi
IoT Platform (e.g. mySQL / MQTT)	Enables real-time communication and data management
Mobile / Web Dashboard	Provides graphical interface for user control and monitoring
Web Browser	Used to access the dashboard remotely
Database	Stores sensor readings, user settings, and lighting history

Chapter III
Design and Development Methodologies

This chapter outlines the specific requirements, features, and functionalities significant to the researchers' system development.

System Design

This section outlines how the Smart Lighting IoT & Energy-Saving System's hardware and software components work together to sense, process, and control environmental lighting. It defines the interaction between LDR and PIR sensors, the Orange Pi microcontroller, relay modules, and the cloud platform to ensure responsive, automated, and energy-efficient operation. This design also enables seamless communication and user-friendly access for both automated control and manual override via the web dashboard.

Architectural Diagram/ Block Diagram

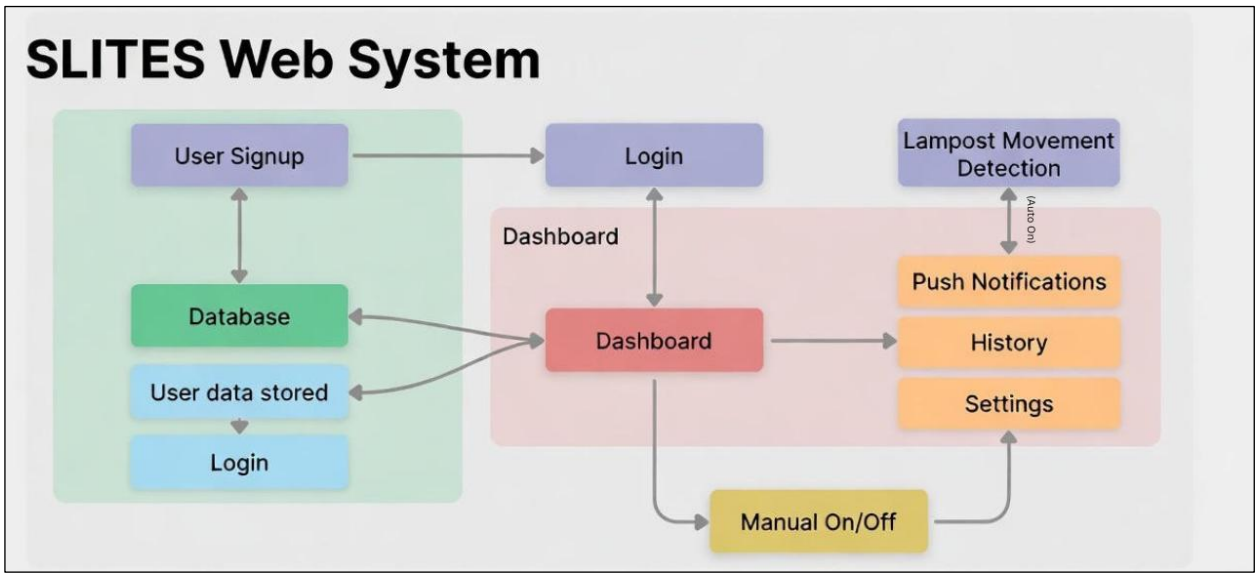


Figure 3 . Architectural Diagram

The diagram represents the flow of data and the interactions between different components within the SLITES smart lamp post

management system. Below is a breakdown of each component and its role in the system:

1. **User Signup**
   - New users access the SLITES website and create an account by entering their details (name, email, password, etc.).
   - Once submitted, the signup information is immediately sent to the Database for secure storage.
   - This step ensures only registered users can access and control the lamp posts.
2. **Login**
   - Registered users enter their email and password on the login page.
   - The system verifies the credentials against the data stored in the Database.
   - Upon successful verification, the user is automatically redirected to the Dashboard. This provides a smooth and secure entry into the system.
3. **Dashboard**
   - The Dashboard is the main control center of SLITES, displayed immediately after login.
   - It shows real-time status of all lamp posts, current light conditions, and quick access to key features.
   - Users can view live data, receive updates, and interact with all controls from this single, easy-to-use screen.
4. **Lamp post Movement Detection**
   - Each smart lamp post is equipped with a motion sensor that constantly monitors for movement.
   - When movement is detected (e.g., a person or vehicle passes by), the sensor automatically turns the light ON.
   - This data is instantly sent to the Dashboard (in real time) and saved in the Database. Users see the light status change immediately without any manual action.
5. **Push Notifications**
   - The system sends instant alerts to the user's browser or device whenever important events occur.
   - Examples: "Light automatically turned ON due to movement", "Light turned OFF manually", or "New activity detected".
   - Notifications appear on the Dashboard and keep users informed even if they are not actively watching the screen.
6. **History**
   - This feature displays a complete log of all lamp post activities.

- Users can view past events such as when lights were turned ON automatically, when they were turned OFF, and who used manual control.
- All history data is retrieved from the Database and presented in a clear, chronological timeline for easy monitoring and reporting.

7. **Settings**
   - Users can personalize the system according to their needs.
   - Options include adjusting notification preferences, setting schedules, changing sensitivity of motion sensors, or managing account details.
   - Any changes made in Settings are saved directly to the Database and take effect immediately across the system.
8. **Manual On/Off**
   - From the Dashboard, users have full control to manually turn any lamp post ON or OFF with a single click04.
   - This feature is useful for overriding automatic mode (e.g., keeping lights ON for security or turning them OFF to save energy).
   - Manual commands are sent instantly to the lamp posts and recorded in both the History and Database.
9. **Database**
   - The Database is the central storage for all information in SLITES.
   - It securely stores user accounts, real-time light status, movement detection records, manual control logs, history data, and user settings.
   - The Dashboard constantly communicates with the Database to display up-to-date information and save every action for reliability and future reference.

Schematic Diagram

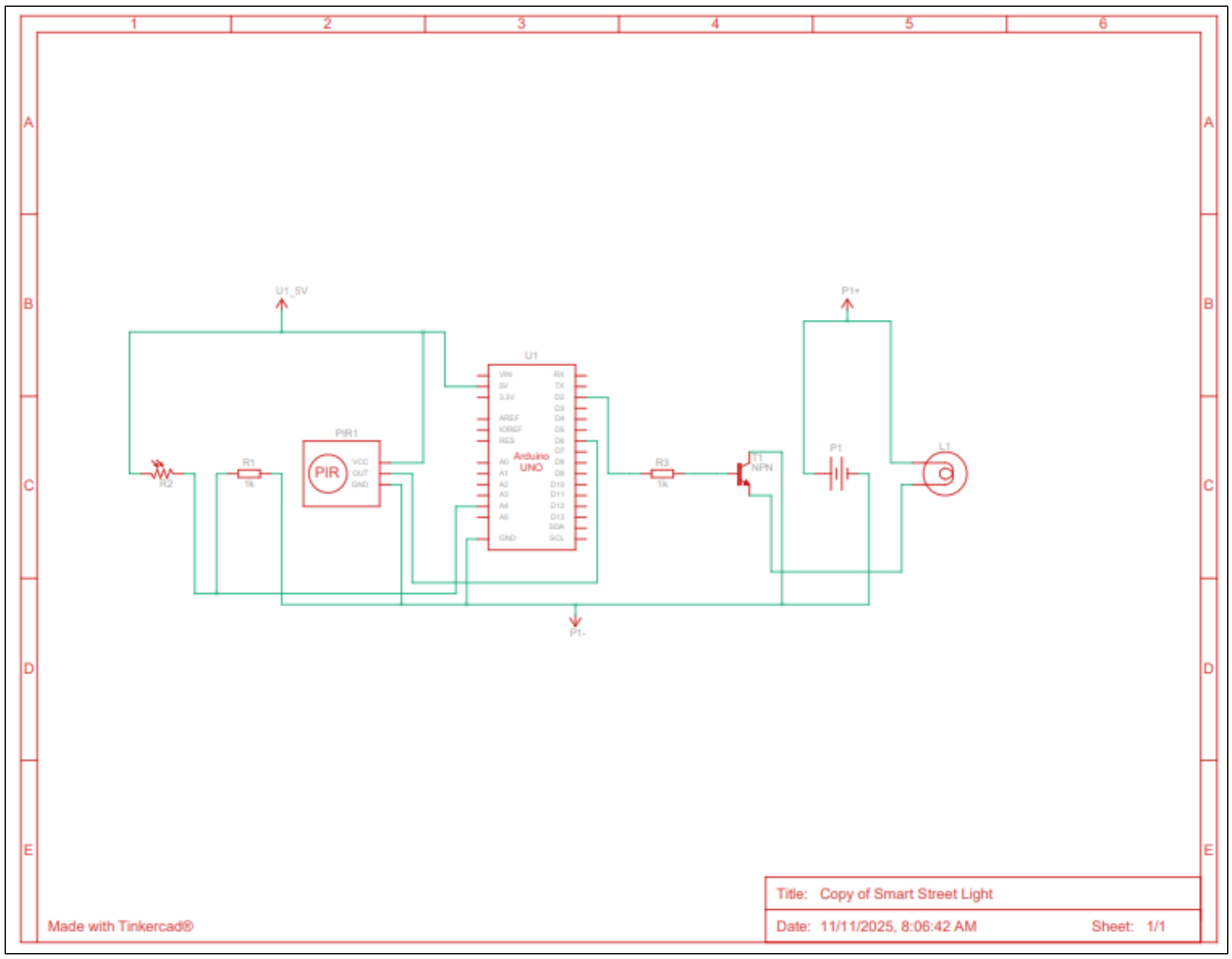


Figure 4. Slites Schematic diagram

The diagram represents the complete electrical connections and signal flow of one SLITES smart lamp post unit. Below is a breakdown of each component and its role in the system:

1. **Power Input (P1+ and Ground)**
   - Main 12V DC supply (usually from solar panel + battery or mains adapter).
   - Provides power to the entire circuit including the high-power lamp and the low-power control section.
2. **5V Voltage Regulator (78L05)**
   - Converts the incoming 12V down to a stable 5V.
   - Supplies clean power to the Arduino UNO and the PIR sensor so they operate safely and reliably.
3. **PIR Motion Sensor (PIR1 – HC-SR501 or similar)**

- Detects movement (people, vehicles) using infrared technology.
- VCC connected to 5V, GND to ground, and OUT connected to Arduino Digital Pin 2.
- When motion is detected, it sends a HIGH signal to the Arduino.

4. **10kΩ Pull-down Resistor**
    - Connected between PIR OUT (Pin 2) and ground.
    - Ensures the input stays LOW when no motion is present, preventing false triggers due to floating voltage.
5. **Arduino UNO (U1)**
    - The microcontroller brain of the lamp post.
    - Pin 2 receives motion signal from PIR (input).
    - Pin 3 sends control signal to the relay (output).
    - Runs simple code: "If motion → turn relay ON for X seconds".
6. **Relay Module**
    - Electrically isolates the low-voltage Arduino from the high-power lamp.
    - Trigger input connected to Arduino Digital Pin 3.
    - Normally Open (NO) contact switches the actual lamp ON/OFF.
    - Contains built-in flyback diode and transistor for safe operation.
7. **Lamp / Light Bulb (L1)**
    - The actual street light (LED, sodium, or any high-power lamp).
    - Connected through the relay's NO and COM terminals.
    - Turns ON only when the relay is activated (either by motion or future manual cloud command).
8. **Ground Connections**
    - All components share a common ground (shown with ↓ symbol).
    - Ensures stable and safe operation of the entire circuit.

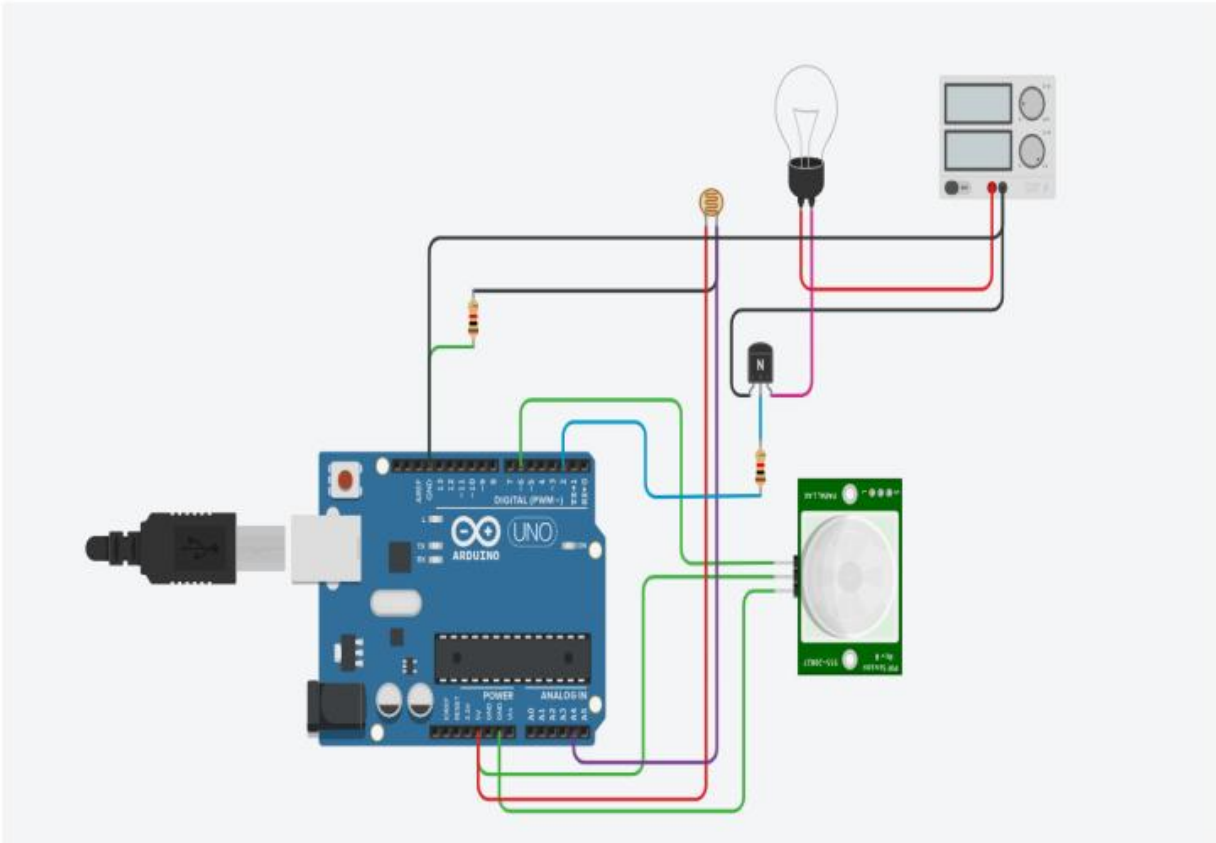


Figure 5. Schematic Diagram

Figure 5 illustrates the system's electronic layout and circuit connections. It shows how the LDR and PIR sensors, Orange Pi, relay module, and power components are wired to enable control and communication. This diagram serves as a technical guide for hardware assembly, ensuring correct power distribution and signal flow between all electronic devices.

3D Model

Figure 6 & 7. 3D Model

Figure 6 & 7 showcase the physical design and form factor of the Smart Lighting Automation System. They present a realistic visualization of the housing, sensor placements, relay module, and wiring layout, providing insight into both functionality and compactness. The model ensures practical assembly, safe component

isolation, and accessibility for maintenance, while also maintaining a durable and tidy enclosure suitable for installation in residential or office environments. It aids in evaluating spatial arrangement, heat dissipation, and ease of access before final fabrication and assembly.

Testing and Evaluation

This section presents the procedures conducted to test and evaluate the SLITES (Smart Lighting Intelligence for Targeted Energy Saving) system throughout its development. It explains the system's major features, the operational functions of each component, and their roles in achieving the overall objectives of the IoT-based smart lighting framework. The testing process also details the operational workflow, ensuring that users and evaluators clearly understand how the system is accessed, controlled, and monitored.

A series of structured test scenarios were executed to verify whether each module—hardware, software, automation logic, and user interface—performs according to the specified functional requirements. These scenarios include user authentication, sensor detection, relay switching, dashboard monitoring, manual override operation, and system logging. Each function was tested under normal and simulated real-world conditions to ensure consistency, accuracy, and responsiveness.

The evaluation further involved user participation through ISO 25010-based assessment, allowing respondents to rate the system across various software quality characteristics such as functionality, performance efficiency, usability, reliability,

security, maintainability, and portability. The results from these tests and evaluations provide comprehensive evidence of the system's stability, correctness, and effectiveness prior to its potential deployment in real-world environments. The findings confirm that SLITES performs reliably as a smart lighting solution and meets the intended goals of automated, energy-efficient illumination management.

ISO 25010 Evaluation

This section presents the evaluation of the system using the ISO 25010 Software Quality Model. It focuses on assessing the system's overall quality based on user-experienced factors such as functionality, reliability, usability, and performance efficiency. The gathered feedback helps determine how well the system meets user needs and performs in real usage.

Table 6. Functionality of the System

Functionality	Mean	Rank	Verbal Interpretation
1.1 The system accurately detects ambient light levels using the LDR sensor.	3.5	1	Strongly Agree
1.2 The PIR motion sensor reliably triggers	3.4	3	Agree

lighting activation.			
1.3 The automation logic correctly turns lights ON/OFF based on sensor input.	3.3	5	Agree
1.4 The remote control feature via the web dashboard works consistently.	3.4	3	Agree
1.5 The manual override feature functions as intended without disrupting automation.	3.5	1	Strongly Agree
Overall mean	**3.42**		**Agree**

Table 6 presents the respondents' evaluation of the system's functionality. Both ambient light detection (1.1) and manual override (1.5) received the highest mean (3.5, Rank 1), indicating users found these features highly reliable. Motion-triggered activation (1.2) and remote control (1.4) followed closely with a mean of 3.4 (Rank 3). The automation logic (1.3) received the

lowest rating (3.3, Rank 5), suggesting minor opportunities for fine-tuning. Overall, the system's functionality is positively rated with a mean of 3.42, reflecting that it meets user expectations for core performance.

Table 7. Reliability of the System

Reliability	Mean	Rank	Verbal Interpretation
2.1 The system operates without unexpected crashes or disconnections.	3.1	4	Agree
2.2 Sensor readings remain stable and consistent over repeated use.	3.6	1	Strongly Agree
2.3 The system functions correctly under various environmental conditions (e.g., changing light, motion patterns).	3.4	3	Agree
2.4 Cloud data logging and	3.5	2	Strongly Agree

synchronization work reliably.			
2.5 The hardware (Orange Pi, relay) operates reliably during prolonged use.	3.0	5	Agree
Overall mean	**3.32**		**Agree**

Table 7 presents the respondents' evaluation of the system's reliability. Sensor reading consistency (2.2) received the highest mean (3.6, Rank 1), showing strong confidence in data stability. Cloud data logging (2.4) followed with a mean of 3.5 (Rank 2). Performance under varying conditions (2.3) and system stability (2.1) were also positively rated. The lowest score was for prolonged hardware reliability (2.5, 3.0, Rank 5), indicating some concerns over long-term durability. Overall, the system demonstrates good reliability with a mean of 3.32.

Table 8. Usability of the System

Usability	Mean	Rank	Verbal Interpretation
3.1 The web dashboard interface is easy to understand and navigate.	3.7	1	Strongly Agree
3.2 Dashboard controls	3.6	2	Strongly Agree

(ON/OFF, settings) are clear and intuitive.			
3.3 Users can monitor lighting status and sensor data without assistance.	3.5	3	Strongly Agree
3.4 The system settings are easy to configure (e.g., adjusting light thresholds).	3.3	5	Agree
3.5 Notifications (e.g., motion detected, light turned on) are clear and useful.	3.4	4	Agree
Overall mean	**3.50**		**Strongly Agree**

Table 8 presents the respondents' evaluation of the system's usability. The web dashboard interface (3.1) received the highest

mean (3.7, Rank 1), indicating users found it very easy to navigate. Dashboard controls (3.2) and independent monitoring (3.3) also scored highly. Configuring system settings (3.4) received the lowest rating (3.3, Rank 5), suggesting this aspect could be simplified. Overall, usability is rated very positively with a mean of 3.50, reflecting a strong and user-friendly experience.

Table 9. Efficiency of the System

Efficiency	Mean	Rank	Verbal Interpretation
4.1 The system responds quickly to motion detection.	3.6	1	Strongly Agree
4.2 The delay between sensor input and light activation is minimal.	3.5	2	Strongly Agree
4.3 The dashboard loads and updates data with minimal delay.	3.3	4	Agree
4.4 Switching between manual and automated	3.2	5	Agree

modes is smooth and fast.			
4.5 The system operates without consuming excessive network or processing resources.	3.4	3	Agree
Overall mean	**3.40**		**Agree**
Efficiency	Mean	Rank	Verbal Interpretation

Table 9 presents the respondents' evaluation of the system's efficiency. Quick motion response (4.1) received the highest mean (3.6, Rank 1), and minimal activation delay (4.2) followed closely (3.5, Rank 2), showing the system is perceived as responsive. Resource efficiency (4.5) and dashboard loading (4.3) were also rated positively. The lowest score was for mode switching (4.4, 3.2, Rank 5), indicating a slight delay in this function. Overall, the system's efficiency is well-regarded with a mean of 3.40, reflecting effective and timely performance.

Respondents Profile

The respondents of this study will be selected from third-year and fourth-year BSIT students with prior coursework in IoT systems or embedded programming. These participants were chosen due to their technical background, enabling them to meaningfully evaluate the SLITES system during hands-on testing. Their insights

will support the assessment of the system's functionality, reliability, usability, and efficiency from a well-informed perspective.

Table 10. Respondents of the Study

Respondents	Number of Respondents	Percentage
Third-year BSIT Students	5	50%
Fourth-year BSIT Students	5	50%
Total	**10**	**100%**

Table 10 presents the classification of the respondents involved in this study, showing the distribution of participants across the two year levels. The table includes the total number and percentage of third-year and fourth-year BSIT students who took part in the evaluation. This balanced distribution ensures a representative sample for assessing the SLITES system's performance from users with foundational to intermediate technical knowledge.

Likert Scale

The researchers surveyed the selected respondents using a Likert Scale questionnaire to evaluate the project's effectiveness. This assessment measured users' perceptions of the system's performance across key quality attributes, providing structured insights into its practical operation and user experience.

Table 11. Four-Point Likert Scale

Scale	Range	Verbal Interpretation
4	3.50–4.00	Strongly Agree
3	2.50–3.49	Agree
2	1.50–2.49	Disagree
1	1.00–1.49	Strongly Disagree

Table 11 outlines the Likert Scale utilized to evaluate the survey responses. It classifies feedback into four levels, from "Strongly Agree" to "Strongly Disagree," each assigned a corresponding numerical value. This approach allows for the systematic measurement of participants' agreement, offering a clear and consistent method for interpreting the collected data.

Evaluation Instrument

To assess the overall performance, quality, and user satisfaction of the **SLITES**, the researchers utilized a structured survey questionnaire based on the **ISO 25010 Software Quality Model**. A four-point Likert Scale was used to measure the respondents' level of agreement with statements evaluating key quality characteristics: **functionality**, **reliability**, **usability**, **and efficiency**.

The survey served as the primary evaluation instrument, allowing users to provide feedback after directly interacting with the system's hardware (LDR, PIR, relay, Orange Pi) and software (dashboard, automation logic, remote control). Selected third-year and fourth-year BSIT students were asked to test the system in a controlled environment and rate their experience during operation. Their responses provided quantitative data that were analyzed to

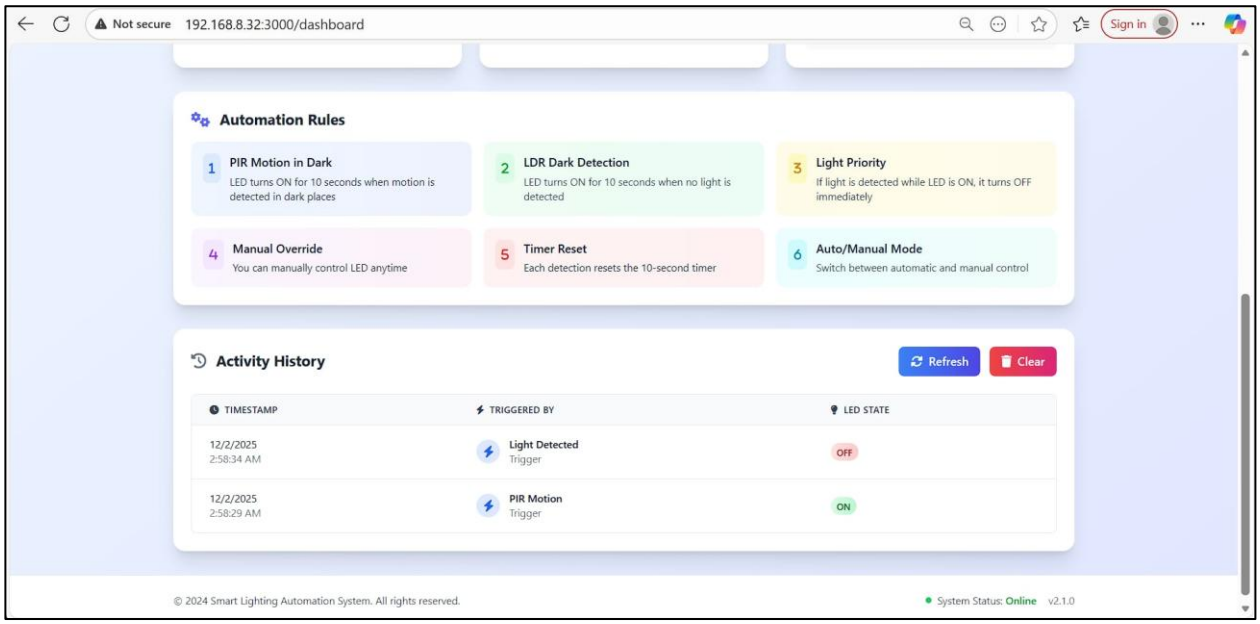
determine the system's strengths and identify potential improvements.

Through this method, the researchers ensured that the evaluation was systematic, measurable, and aligned with international software quality standards, supporting the validity and reliability of the study's findings.

CHAPTER IV
DEVELOPMENT, TESTING AND EVALUATION RESULT

Presentation of the System Output

The SLITES: Smart Lighting Intelligence for Targeted Energy Saving system produced several functional outputs that demonstrate its capability as an IoT-based automated lighting solution. This section presents the major system components, interfaces, and functionalities developed throughout the study.



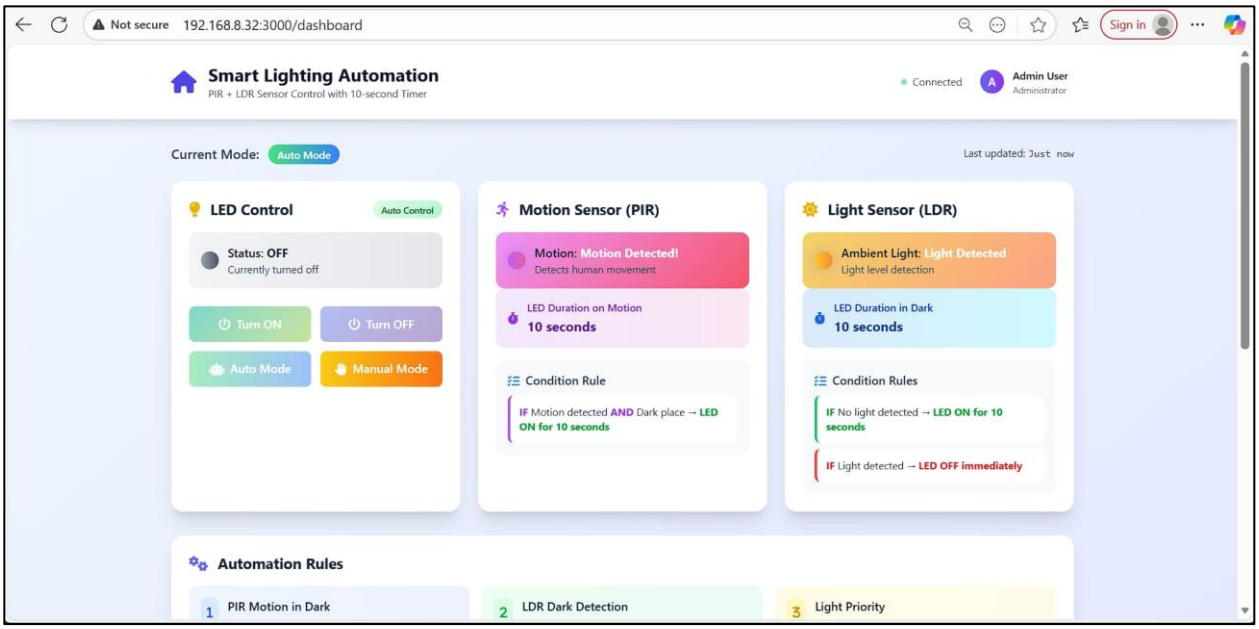


Figure 8 & 9. Dashboard

1. User Authentication Module

The system includes a secure Login and Signup feature that allows users to create an account and access the system using valid credentials. This ensures that only authorized users can monitor and control the smart lamp posts. The authentication process verifies user identity and redirects them to the dashboard upon successful login.

2. Real-Time Monitoring Dashboard

The Dashboard serves as the main interface where users can view and manage all lamp posts in real time. It displays:

- Current lamp status (ON/OFF)
- Sensor activity such as detected movement
- Timestamped events and notifications
- Quick-access buttons for manual control

The dashboard updates automatically as sensor data is sent from the device to the server.

3. Motion-Based Light Automation

The system integrates a PIR motion sensor that automatically detects movement. When activity is sensed:

- The lamp post turns ON instantly.
- The event is recorded in the system history.
- A dashboard notification is generated.

If no movement is detected after a set duration, the system automatically switches the light **OFF**, promoting energy efficiency.

4. Manual ON/OFF Override Control

Aside from automation, users can manually turn lamp posts ON or OFF through the dashboard. This feature is essential for emergency use, maintenance, or special conditions where manual intervention is required. Actions performed manually are also logged for monitoring.

5. System Settings Module

Users can personalize the system through the Settings page. Adjustable options include:

- Motion detection sensitivity
- Notification preferences
- User information updates
- System behavior configurations

All settings are stored in the database and applied system-wide.

6. Activity History Logs

The History module provides a complete log of system activities, including:

- Automated light activations
- Manual ON/OFF controls
- Detected movements
- System events and timestamps

This supports transparency, traceability, and system monitoring.

7. Hardware Output (Prototype Lamp Post Unit)

The hardware prototype demonstrates successful integration of:

- PIR motion sensor for movement detection
- Relay module for switching the lamp
- Microcontroller for processing
- LED lamp as the lighting output

When motion is detected, the lamp switches ON. When the environment is idle, it switches OFF according to programmed conditions. All hardware activities synchronize with the dashboard in real time.

8. Notification System

The system generates automated push-style notifications for significant events such as:

- Movement detected
- Light turned ON or OFF
- Manual override activation

Notifications help users stay informed even without actively monitoring the dashboard.

Testing Results

The testing results provide the outcomes of the system evaluations conducted to assess the SLITES system's functionality, responsiveness, and operational stability. They summarize findings from sensor performance verification, automation logic accuracy, remote control effectiveness, and system behavior under varying environmental conditions. The results confirm the extent to which the system meets its defined technical and user experience requirements.

ISO 25010 Evaluation Result

This section presents the evaluation of the SLITES system using the ISO 25010 Software Quality Model. It focuses on assessing the system’s overall quality based on user-experienced factors such as functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The collected feedback determines how well the system meets user needs and performs during actual usage.

The SLITES smart lighting system was evaluated using the ISO/IEC 25010 Software Quality Model to measure its overall quality based on eight characteristics. A total of respondents rated each statement using a 4-point Likert Scale:

4 – Strongly Agree (SA)

3 – Agree (A)

2 – Disagree (D)

1 – Strongly Disagree (SD)

The following tables present the consolidated summary and analysis based on the responses.

Table 12. Functionality Suitability of the System

Functionality Suitability	Mean	Rank	Verbal Interpretation
1.1. The system performs its intended functions correctly.	3.19	3	Agree
1.2. The outputs of the system are complete and accurate.	3.00	5	Agree
1.3. The system features meet the required objectives.	3.13	4	Agree
Overall mean	**3.11**		**Agree**

Table 12 presents the evaluation of the system's functional suitability. The highest-rated item is the system's ability to perform its intended functions correctly (3.19, Rank 3), showing that respondents generally agree that SLITES fulfills its core purpose of automated lighting control and real-time monitoring. The completeness and accuracy of outputs (3.00, Rank 5) received the lowest mean, indicating that while the system performs well, minor refinement in data presentation or responsiveness may further enhance accuracy. Overall, the functional suitability is rated positively with an overall mean of 3.11, demonstrating that SLITES effectively meets its intended objectives.

Table 13. Performance Efficiency of the System

Performance Efficiency	Mean	Rank	Verbal Interpretation
2.1 The system responds quickly during operation.	3.18	2	Agree
2.2 Sensor data is processed without delay.	3.13	3	Agree
2.3 Pages, modules, and dashboard load efficiently.	3.31	1	Agree
Overall mean	**3.27**		**Agree**

Table 13 presents the respondents' evaluation of performance efficiency. The dashboard's efficient loading time received the highest mean (3.31, Rank 1), suggesting responsive transitions between pages and modules. The system's quick response during operation (3.38, Rank 2) and real-time sensor processing (3.13, Rank 3) indicate consistent performance with minimal delay. The overall mean of 3.27 reflects that SLITES performs efficiently and meets user expectations in real-time operation.

Table 14. Compatibility of the System

Compatibility	Mean	Rank	Verbal Interpretation
3.1 The system works properly with the hardware components.	3.31	1	Agree
3.2 The system functions correctly across different devices.	3.25	2	Agree
Overall mean	**3.28**		**Agree**

Table 14 shows that compatibility received strong positive evaluations. The system's integration with hardware components (3.31, Rank 1) demonstrates stable communication between the microcontroller, sensors, and relay. Its consistent performance across various devices (3.25, Rank 2) highlights cross-platform usability. Overall, the compatibility mean of 3.28 confirms that SLITES operates smoothly in different environments and hardware setups.

Table 15. Usability of the System

Usability	Mean	Rank	Verbal Interpretation
4.1 The interface is easy to understand and use.	3.31	3	Agree
4.2 System navigation is simple and user-friendly.	3.38	1	Agree
4.3 The system can be used with minimal instructions.	3.31	3	Agree
Overall mean	**3.33**		**Agree**

Table 15 presents the usability results. System navigation received the highest rating (3.38, Rank 1), indicating that users found the dashboard intuitive and well-organized. The ease of understanding the interface (3.31, Rank 3) and minimal need for instructions (3.31, Rank 3) demonstrate user-friendliness. The overall mean of 3.33 reflects that SLITES offers a clear and accessible interface suitable for both first-time and regular users.

Table 16. Reliability of the System

Reliability	**Mean**	**Rank**	**Verbal Interpretation**
5.1 The system operates consistently without errors..	3.31	1	Agree
5.2 The system is stable even during continuous use.	3.25	2	Agree
Overall mean	**3.28**		**Agree**

Table 16 presents the reliability evaluation. The system's consistent error-free operation (3.31, Rank 1) was highly rated, indicating stable performance during extended usage. Continuous operation (3.25, Rank 2) also demonstrates overall reliability. The overall mean of 3.28 confirms that SLITES maintains dependable operation and does not encounter significant system failures during testing.

Table 17. Security of the System

Security	**Mean**	**Rank**	**Verbal Interpretation**
6.1 The login process is secure.	3.31	2	Agree
6.2 User data is protected from unauthorized access.	3.38	1	Agree
Overall mean	**3.34**		**Agree**

Table 17 indicates that security was positively evaluated. The protection of user data (3.38, Rank 1) received the highest score, highlighting strong user confidence in the system's data security measures. The login security (3.31, Rank 2) also shows that SLITES provides safe access control. The overall mean of 3.34 suggests that the system effectively implements essential security

protocols.

Table 18. Maintainability of the System

Maintainability	**Mean**	**Rank**	**Verbal Interpretation**
7.1 The system is easy to update or modify.	3.38	1	Agree
7.2 Troubleshooting and system maintenance can be performed easily.	3.31	2	Agree
Overall mean	**3.34**		**Agree**

Table 18 shows that maintainability received positive evaluations. The ease of updating and modifying the system (3.38, Rank 1) indicates that the system's modular design supports future expansion. Troubleshooting and maintenance (3.31, Rank 2) were also rated highly. The overall mean of 3.34 affirms good maintainability and potential for long-term system improvements.

Table 19. Portability of the System

Maintainability	**Mean**	**Rank**	**Verbal Interpretation**
8.1 The system can be deployed in different environments.	3.31	2	Agree
8.2 The system is adaptable to various hardware setups.	3.38	1	Agree
Overall mean	**3.34**		**Agree**

Table 19 demonstrates that portability was rated positively by respondents. Adaptability to different hardware setups (3.38,

Rank 1) and deployment across various environments (3.31, Rank 2) shows the system's flexibility. The overall mean of 3.34 indicates that SLITES is scalable and can be used in diverse installation settings, such as indoor pathways, outdoor lamp posts, or campus walkways.

Respondents Profile

The respondents for the evaluation consisted of BSIT third-year and fourth-year students who interacted directly with the SLITES system during testing. Their feedback provided valuable insights into the functionality, usability, and overall effectiveness of the system.

Table 20. Respondents of the Study

Respondents	Number of Respondents	Percentage
Third-year BSIT Students	5	50%
Fourth-year BSIT Students	5	50%
Total	10	**100%**

Table 20 presents the demographic distribution of the respondents who participated in the evaluation of the SLITES system. A total of 10 BSIT students were involved in the testing process. These students were selected because of their technical background and familiarity with information systems and IoT-related applications, enabling them to provide meaningful and relevant feedback on the system's performance and usability. Their

participation ensured that the evaluation results were informed by individuals with adequate knowledge of software and hardware interaction, supporting a more accurate assessment of the SLITES system's effectiveness. With all respondents actively engaging with the prototype, the gathered data serves as a reliable basis for the overall system evaluation.

Likert Scale

The researchers surveyed with selected respondents using a Likert Scale questionnaire to evaluate the project's effectiveness. This assessment measured the system's impact on users' experiences and perceptions, providing valuable insights into its usability and overall performance.

Table 21. Four-Point Likert Scale

Scale	Range	Verbal Interpretation
4	3.50-4.00	Strongly Agree
3	2.50-3.49	Agree
2	1.50-2.49	Disagree
1	1.00-1.49	Strongly Disagree

Table 21 outlines the Likert Scale utilized to evaluate the survey responses. It classifies feedback into four levels, from "Strongly Agree" to "Strongly Disagree," each assigned with a corresponding numerical value. This approach allows for the systematic measurement of participants' perceptions and experiences, offering a structured way to interpret the collected data.

The overall ISO 25010 evaluation shows that the SLITES system performs effectively across all major software quality categories, receiving predominantly Agree and Strongly Agree ratings from respondents. With no "Strongly Disagree" responses and only minimal "Disagree" feedback under Functional Suitability, the results confirm that the system is functionally accurate, performs efficiently, ensures hardware compatibility, offers excellent usability, and maintains strong reliability and security. Furthermore, its maintainability and portability demonstrate readiness for future upgrades and broader deployment. Overall, the evaluation validates SLITES as a well-designed and effective IoT smart lighting solution suitable for real-world use.

To assess the overall performance, quality, and user satisfaction of SLITES: Smart Lighting Intelligence for Targeted Energy Saving, the researchers utilized a structured survey questionnaire based on the ISO 25010 Software Quality Model. A four-point Likert scale was employed to measure respondents' level of agreement with statements evaluating key quality characteristics such as functionality, reliability, usability, performance efficiency, security, maintainability, compatibility, and portability. The survey served as the primary evaluation instrument, allowing users to provide feedback after interacting with the system. Selected third-year and fourth-year BSIT students were asked to operate the system and rate their experience during its use. Their responses provided quantitative data that were analyzed to determine the system's strengths and identify areas for improvement. Through this systematic and measurable survey method, the researchers ensured that the evaluation was aligned with international

software quality standards, supporting the validity and reliability of the study's findings.

CHAPTER V

CONCLUSION AND RECOMMENDATION

Conclusion

The development of SLITES: Smart Lighting Intelligence for Targeted Energy Saving successfully demonstrated the potential of IoT-based automation in optimizing energy consumption through intelligent lighting control. Using the Orange Pi microcontroller, PIR motion sensors, LDR sensors, and a relay-controlled lighting

output, the system achieved real-time automation and remote accessibility through its dashboard. The integration of hardware and software worked effectively, enabling automated switching, manual override options, activity logging, and user-managed settings. Based on the ISO 25010 evaluation, the system consistently received Agree ratings across all eight software quality characteristics—functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Respondents affirmed that the system performs its intended functions, responds efficiently, works across different devices, and remains user-friendly and reliable. Security, maintainability, and adaptability also scored positively, indicating strong potential for future scaling and enhancements. Overall, the study concludes that SLITES is a functional and efficient prototype capable of supporting energy-saving initiatives through automated, sensor-based lighting control. It proves that IoT technology can effectively contribute to sustainability efforts by reducing unnecessary power usage while maintaining user convenience and system accessibility.

Recommendation

Based on the findings and evaluation results, several recommendations are proposed to further enhance SLITES and prepare it for wider and more advanced deployment: Improve Sensor Calibration and Accuracy Fine-tuning of PIR and LDR sensors is recommended to increase detection reliability, especially for outdoor environments or long-range applications. Weatherproofing and stronger sensor modules may be considered for future versions. Enhance Dashboard Features Additional functionalities such as graphical data visualization, energy consumption reports, and

customizable automation schedules can improve user experience and system monitoring. Integrate More Advanced IoT Capabilities Future development may include features like voice command control, AI-driven adaptive lighting, geo-fencing activation, and integration with smart home ecosystems. Strengthen Security Measures Implementing multi-factor authentication, encrypted data logs, and enhanced user-role permissions will help protect sensitive user information and system activity. Expand Hardware Compatibility Testing the system with various microcontrollers (ESP32, Raspberry Pi) and different sensor types can broaden deployment options and improve cost-efficiency. Develop a Full Outdoor/Commercial Version For real-world use such as street lighting or public pathway illumination, it is recommended to upgrade the hardware for higher durability, weather resistance, and solar-powered operation. Conduct Large-Scale Testing Wider testing with more respondents and in different environments will provide more accurate data on performance, efficiency, and system limitations.

REFERENCES

Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M., & Ayyash, M. (2015). Internet of Things: A survey on enabling technologies, protocols, and applications. *IEEE Communications Surveys & Tutorials, 17*(4), 2347–2376. https://doi.org/10.1109/COMST.2015.2444095

Atzori, L., Iera, A., & Morabito, G. (2010). The Internet of Things: A survey. *Computer Networks, 54*(15), 2787–2805. https://doi.org/10.1016/j.comnet.2010.05.010

Blynk Inc. (2024). *Blynk documentation: Getting started with IoT projects*. Retrieved from https://docs.blynk.io/

Gubbi, J., Buyya, R., Marusic, S., & Palaniswami, M. (2013). Internet of Things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems, 29*(7), 1645–1660. https://doi.org/10.1016/j.future.2013.01.010

Khan, M. A., & Salah, K. (2018). IoT security: Review, blockchain solutions, and open challenges. *Future Generation Computer Systems, 82*, 395–411. https://doi.org/10.1016/j.future.2017.11.022

Moreno, M. V., Úbeda, B., Skarmeta, A. F., & Zamora, M. A. (2014). How can we tackle energy efficiency in IoT based smart buildings? *Sensors, 14*(6), 9582–9614. https://doi.org/10.3390/s140609582

MQTT Organization. (2023). *MQTT version 5.0 specification*. OASIS Standard. https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html

Perera, C., Zaslavsky, A., Christen, P., & Georgakopoulos, D. (2014). Sensing as a service model for smart cities supported by Internet of Things. *Transactions on Emerging Telecommunications Technologies, 25*(1), 81–93. https://doi.org/10.1002/ett.2704

Shah, S. H., & Yaqoob, I. (2016). A survey: Internet of Things (IoT) technologies, applications and challenges. In *2016 IEEE Smart Energy Grid Engineering (SEGE)* (pp. 1–6). IEEE. https://doi.org/10.1109/SEGE.2016.7589532

Zanella, A., Bui, N., Castellani, A., Vangelista, L., & Zorzi, M. (2014). Internet of Things for smart cities. *IEEE Internet of Things Journal, 1*(1), 22–32. https://doi.org/10.1109/JIOT.2014.2306328

Orange Pi Development Board Documentation. (2024). *Orange Pi official hardware and software guide*. Retrieved from http://www.orangepi.org/

Arduino Official Documentation. (2024). *Arduino UNO Rev3 technical reference*. Retrieved from https://docs.arduino.cc/hardware/uno-rev3

International Organization for Standardization. (2011). *ISO/IEC 25010:2011 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models*. ISO.

Delea, F., & Casazza, M. (2020). *Smart lighting systems: A review of control strategies and energy-saving potential*. *Energy Reports, 6*, 132–140. https://doi.org/10.1016/j.egyr.2020.11.098

Patil, S., & Patil, S. (2021). IoT-based smart street light system using PIR and LDR sensors with energy-saving approach. *International Journal of Advanced Research in Computer and Communication Engineering, 10*(5), 97–103.

APPENDICES

Sample Accomplished ISO 25010 Evaluation Form

ISO 25010 Evaluation

Respondents Profile
Likert Scale
Evaluation Instrument

The respondents of this evaluation consist of individuals who have relevant knowledge of information systems and user interaction. They include:

- Information Technology students
- Faculty members of the College of Computer Studies
- Technical personnel with background in system development

These respondents were selected to provide reliable feedback on the quality and usability of the SLITES: IoT-Based Smart Lighting Intelligence for Targeted Energy Saving system.

Likert Scale

The ISO 25010 evaluation uses a **4-point Likert Scale** to measure the level of agreement of each respondent with the statements provided in the evaluation instrument.

Scale	Description
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

Respondents place a **check mark (✓)** on the box that represents their answer.

Evaluation Instrument

The instrument used for evaluation is based on the **ISO 25010 Software Quality Model,** which measures eight (8) major quality characteristics of a system. Each characteristic includes several statements that respondents will evaluate using checkboxes.

ISO 25010 Evaluation Checklist (Template)

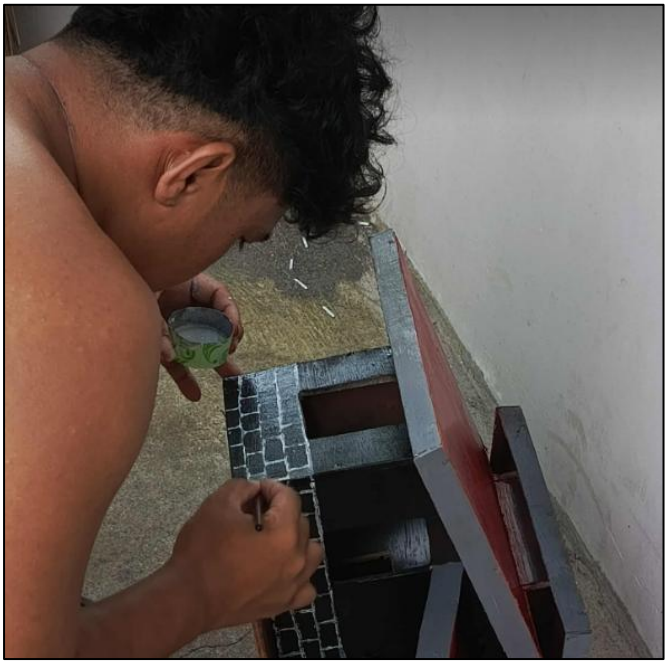
No.	Quality Characteristic / Statement	4	3	2	1
Functional Suitability					
1	The system performs its intended functions correctly.	☐	☐	☐	☐

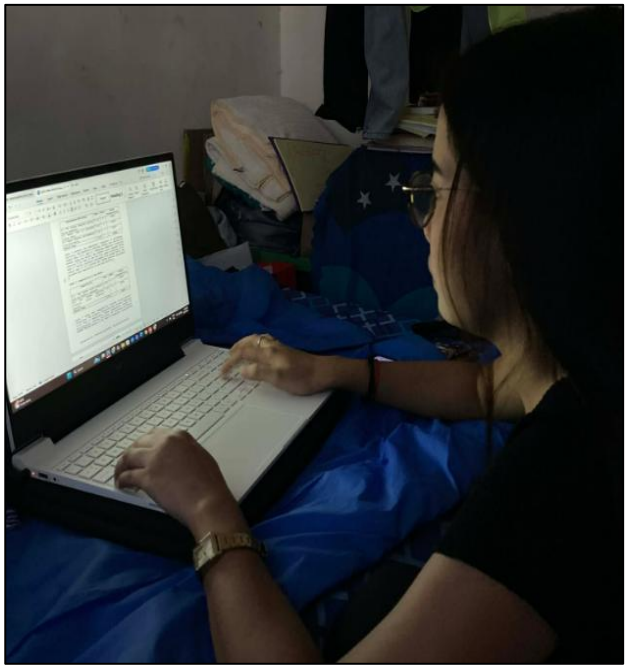
No.	Quality Characteristic / Statement	4	3	2	1
2	The outputs of the system are complete and accurate.	☐	☐	☐	☐
3	The system features meet the required objectives.	☐	☐	☐	☐
Performance Efficiency					
4	The system responds quickly during operation.	☐	☐	☐	☐
5	Sensor data is processed without delay.	☐	☐	☐	☐
6	Pages, modules, and dashboard load efficiently.	☐	☐	☐	☐
Compatibility					
7	The system works properly with the hardware components.	☐	☐	☐	☐
8	The system functions correctly across different devices.	☐	☐	☐	☐
Usability					
9	The interface is easy to understand and use.	☐	☐	☐	☐
10	System navigation is simple and user-friendly.	☐	☐	☐	☐
11	The system can be used with minimal instructions.	☐	☐	☐	☐
Reliability					
12	The system operates consistently without errors.	☐	☐	☐	☐
13	The system is stable even during continuous use.	☐	☐	☐	☐
Security					
14	The login process is secure.	☐	☐	☐	☐
15	User data is protected from unauthorized access.	☐	☐	☐	☐
Maintainability					
16	The system is easy to update or modify.	☐	☐	☐	☐
17	Troubleshooting and maintenance can be performed easily.	☐	☐	☐	☐
Portability					

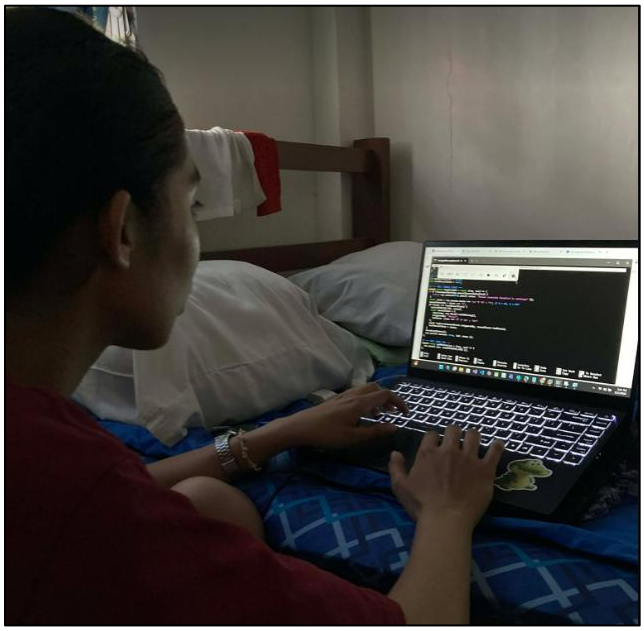
No.	Quality Characteristic / Statement	4 3 2 1
18	The system can be deployed in different environments.	☐☐☐☐
19	The system is adaptable to various hardware setups.	

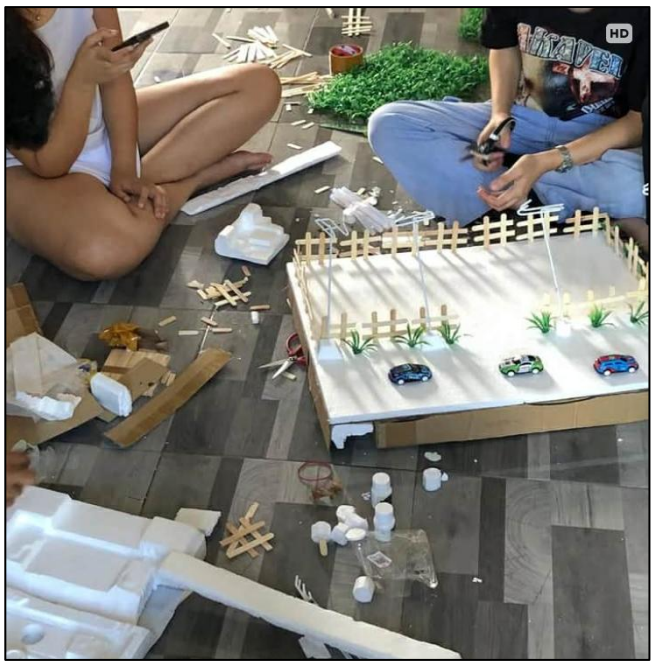
https://docs.google.com/forms/d/e/1FAIpQLSdrtZHKQn9EvpbEGL9bVUa6GTNhdFxtP7lmQUgzfYhpUut9ww/viewform?usp=dialog

Picture During Development, Testing & Evaluation





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CURRICULUM VITAE

Name: Chynna C. Alemania
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PERSONAL DATA

Nickname: Chie
Age: 21 years old
Sex: Female
Birthdate: June 20,2004
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
Language: Tagalog/English
Height: 5'2
Weight: 50kg
Father: Felix Suarez Alemania
Mother: Annie Bool Cabatay

EDUCATION

Elementary: Silonay Elementary School
School Year: 2010-2016
Secondary: Parang National High School
School Year: 2016-2020
Senior High School: Holy Infant Academy
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

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PERSONAL DATA

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Birthdate: February 11,2004
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
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Father: Gaudencio R. Basconcillo, Jr.
Mother: Gemma M. Basconcillo

EDUCATION

Elementary: Parang Elementary School
School Year: 2009-2016
Secondary: Parang National High School
School Year: 2016-2020
Senior High School: Holy Infant Academy
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

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Birthdate: September 08,2003
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
Language: Tagalog/English
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Weight: 50kg
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Mother: Elma G. Mendoza

EDUCATION

Elementary: Parang Elementary School
School Year: 2009-2016
Secondary: Parang National High School
School Year: 2016-2020
Senior High School: Holy Infant Academy
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
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PERSONAL DATA

Nickname: Kitkit
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Sex: F
Birthdate: January 29, 2004
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Citizenship: Filipino
Religion: Protestant
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Weight: 53kg
Father: Neil L. Icalla
Mother: Prescila U. Icalla

EDUCATION

Elementary: Minas Elementary School
School Year: 2009-2016
Secondary: MinSCAT Laboratory High School
School Year: 2016-2020
Senior High School: Good Shepherd Academy
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

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PERSONAL DATA

Nickname: Gybrielle
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Sex: F
Birthdate: March 3, 2004
Civil Status: Single
Citizenship: Filipino
Religion: Catholic
Language: Tagalog/English
Height: 170cm
Weight: 59kg
Father: N/A
Mother: Marilou B. Gacilo

EDUCATION

Elementary: Filemon Samaco Memorial School
School Year: 2009-2016
Secondary: Pedro V. Panaligan National Highschool
School Year: 2016-2020
Senior High School: Oriental Mindoro National Highschool
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

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Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
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Height: 157 cm
Weight: 43 kg
Father: Ariel M. Guerra
Mother: Gloria M. Guerra

EDUCATION

Elementary: N. Aboboto Memorial School
School Year: 2009-2016
Secondary: Parang National High School
School Year: 2016-2020
Senior High School: Parang National High School
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

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PERSONAL DATA

Nickname: Dhogz
Age: 21
Sex: Male
Birthdate: November 28, 2004
Civil Status: Single
Citizenship: Filipino
Religion: Catholic
Language: Tagalog/English
Height: 165cm
Weight: 60kg
Father: Dante JR. Tupaz
Mother: None

EDUCATION

Elementary: Minolo Elementary School
School Year: 2009-2016
Secondary: Puerto Galera Academy
School Year: 2016-2020
Senior High School: Puerto Galera National Highschool
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

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PERSONAL DATA

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Sex: Male
Birthdate: June 24, 2001
Civil Status: Single
Citizenship: Filipino
Religion: Born Again
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Height: 160cm
Weight: 73kg
Father: Reginald C. Fesalbon
Mother: Sharon F. Fesalbon

EDUCATION

Elementary: Concepcion Central Elementary School
School Year: 2008-2014
Secondary: Concepcion National High School
School Year: 2014-2018
Senior High School: Holy Infant Academy
School Year: 2018-2020
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

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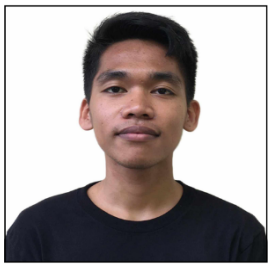
PERSONAL DATA

Nickname: Alvin
Age: 24
Sex: Male
Birthdate: September 28, 2001
Civil Status: Single
Citizenship: Filipino
Religion: INC
Language: Tagalog/English
Height: 162cm
Weight: 61kg
Father: Bautista D. Villaverde Jr.
Mother: Emamie R. Manalo

EDUCATION

Elementary: San Teodoro Central School
School Year: 2006-2014
Secondary: San Teodoro National High School
School Year: 2014-2017
Senior High School: San Teodoro National High School
School Year: 2018-2019
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE



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Sex: Male
Birthdate: August 21, 2001
Civil Status: Single
Citizenship: Filipino
Religion: Christian
Language: Tagalog/English
Height: 163.83cm.
Weight: 56kg.
Father: Charles L. Rabulan
Mother: Reziel M. Rabulan

EDUCATION

Elementary: Ma. Estrella PVG Tawiran Elementary School
School Year: 2006-2012
Secondary: Imus National High School
School Year: 2013-2016
Senior High School: Asian Institute of Computer Studies
School Year: 2017-2019
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate

CURRICULUM VITAE

Name: Mark Gabriel G. Baje
Birthplace: Poblacion, Mariveles Bataan
Contact Number: 0927 846 1256
E-MAIL: bajemarkgabriel263@gmail.com

PERSONAL DATA

Nickname: Gabio
Age: 22
Sex: Male
Birthdate: September 30, 2003
Civil Status: Single
Citizenship: Filipino
Religion: Catholic
Language: Tagalog
Height: 171 cm.
Weight: 81 kg.
Father: Dario C. Baje
Mother: Marivic De Guzman Gipan

EDUCATION

Elementary: Balon Elementary School
School Year: 2009-2016
Secondary: Naujan Municipal Highschool
School Year: 2016-2020
Senior High School: Naujan Municipal Highschool
School Year: 2020-2022
College: Mindoro State University- Calapan Campus
Course: Bachelor of Science in Information Technology
School Year: 2022- Under Graduate